



OBJECT ORIENTED SOFTWARE ENGINEERING

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Advanced relationships in Use Case diagram

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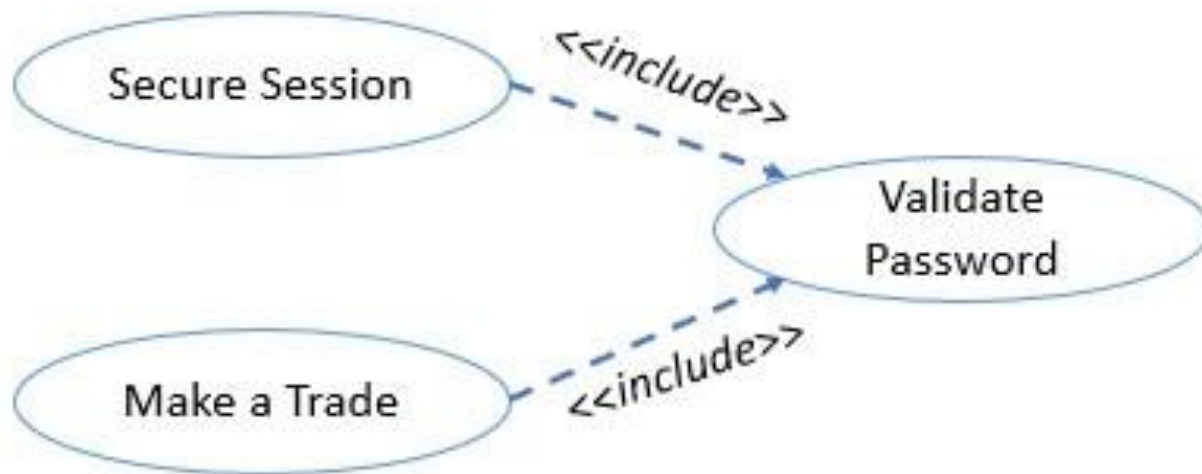
Include Relationship

- The include relationship incorporates **one use case within the behavior sequence of another use case.**
- An included use case is like **a subroutine.**
- The **UML notation** for an included relationship is **dashed arrow from the source use case(including) to the target use case(included).**

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Include Relationship

The keyword `<<include>>` annotates the arrow

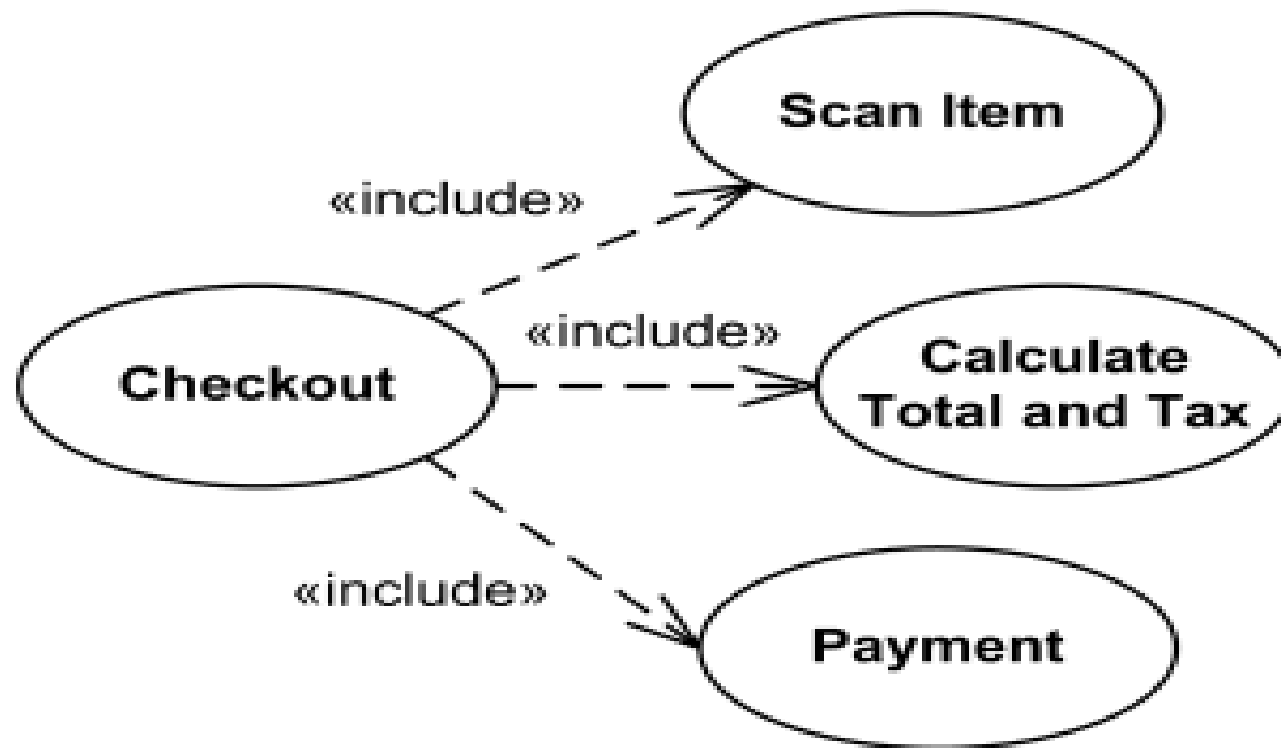


Include Use Case Relationship

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Include Relationship

Another example:



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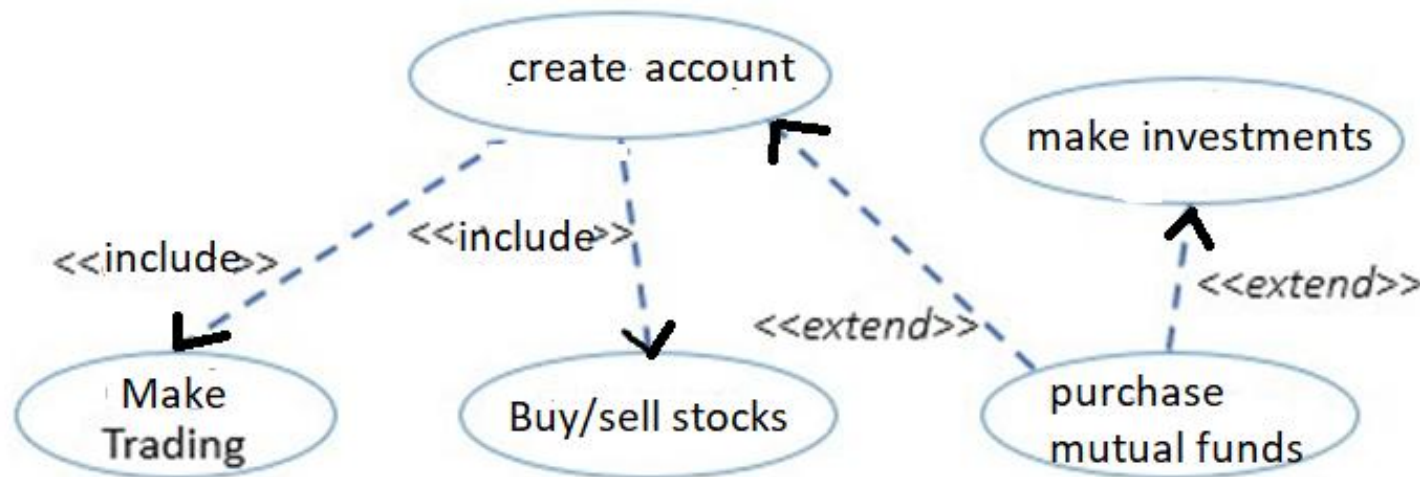
Extended Relationship

- The extended relationship adds incremental behavior to an use case.
- It represents the frequent situation in which some **initial capability** is defined and **later features are added modularity**.

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Extended Relationship ...

- The UML notation for extended relationship is a **dashed arrow from the extension use case to the base use case**.
- The keyword <<extend>> annotates the arrow



Extend Use Case Relationship

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Extended Relationship ...

- The extend relationship **connects** an **extension use case to a base use case**.
- The base use case must be a **valid use case** in the **absence of any extensions**.

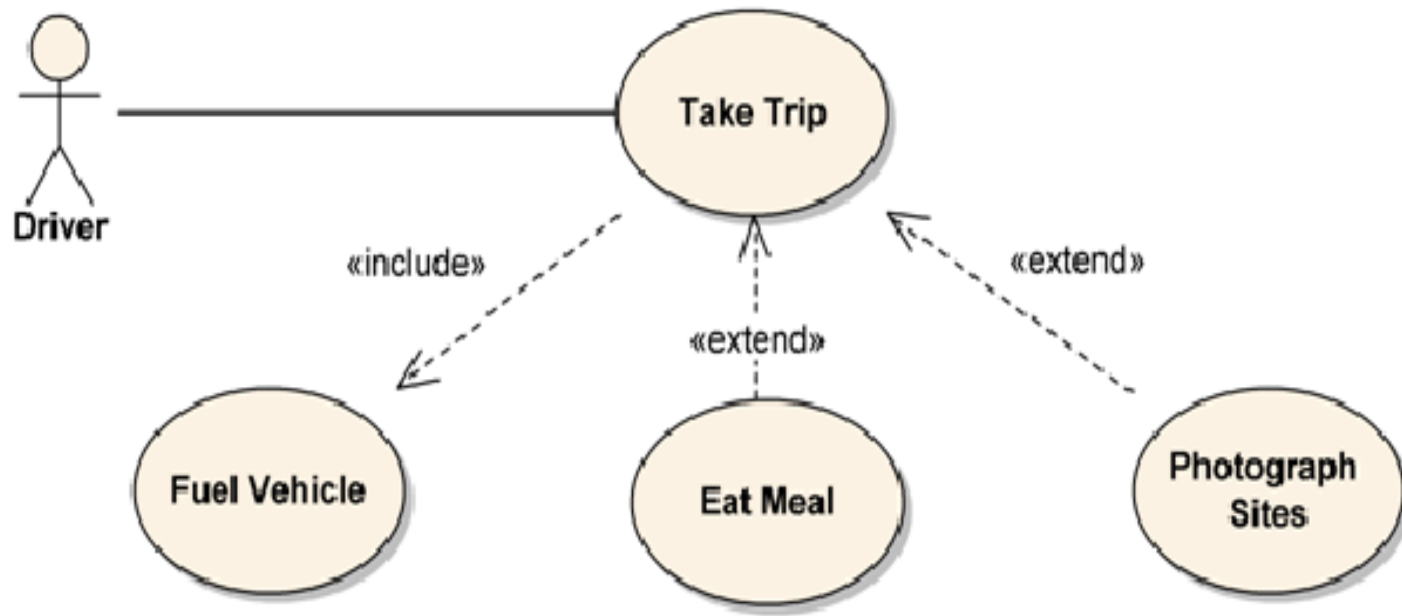
The behavior sequence of the extension use case occurs at the given point in the sequence.

The extension behavior **occurs only if the condition is true** when control reaches the insert location.

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Extended Relationship ...

- Consider another example: Car Rental Application



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3. Use Case Generalization

- **Generalization** is the most common term in computers. In generalization, there is a **parent class** consisting of common attributes and operations and **the child class** contains the refined attributes and operations.
- Analogues to generalization in classes there is a concept of generalization in use case relationship.

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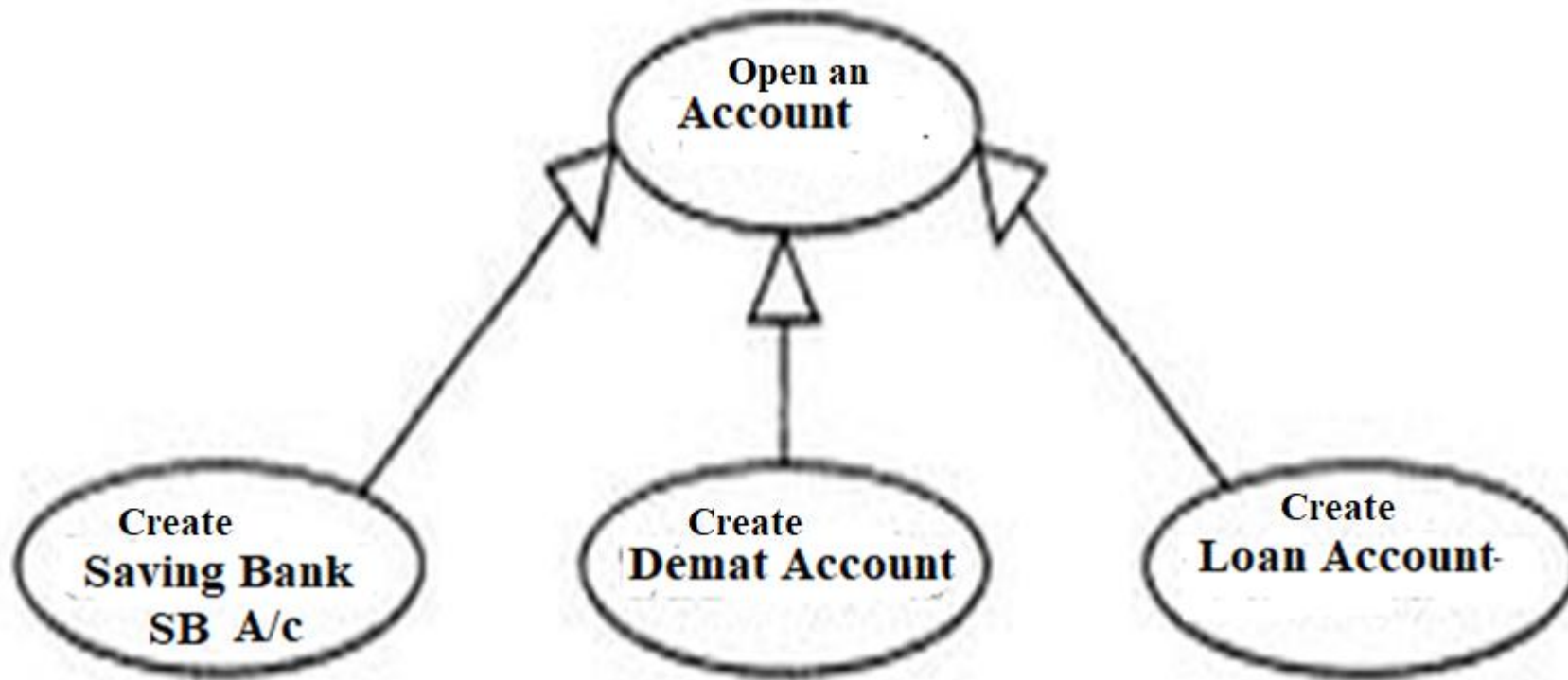
Use Case Generalization

Generalization can **show specific variations** on a general use case.

A **parent use case** represents a **general behavior** sequence. Child use case **specialize** the parent by inserting **additional steps** or by **refining steps**.

The UML indicates generalization by **arrow** with its **tail on the child use case** and a **triangular arrow head on the parent use case**.

For Example:



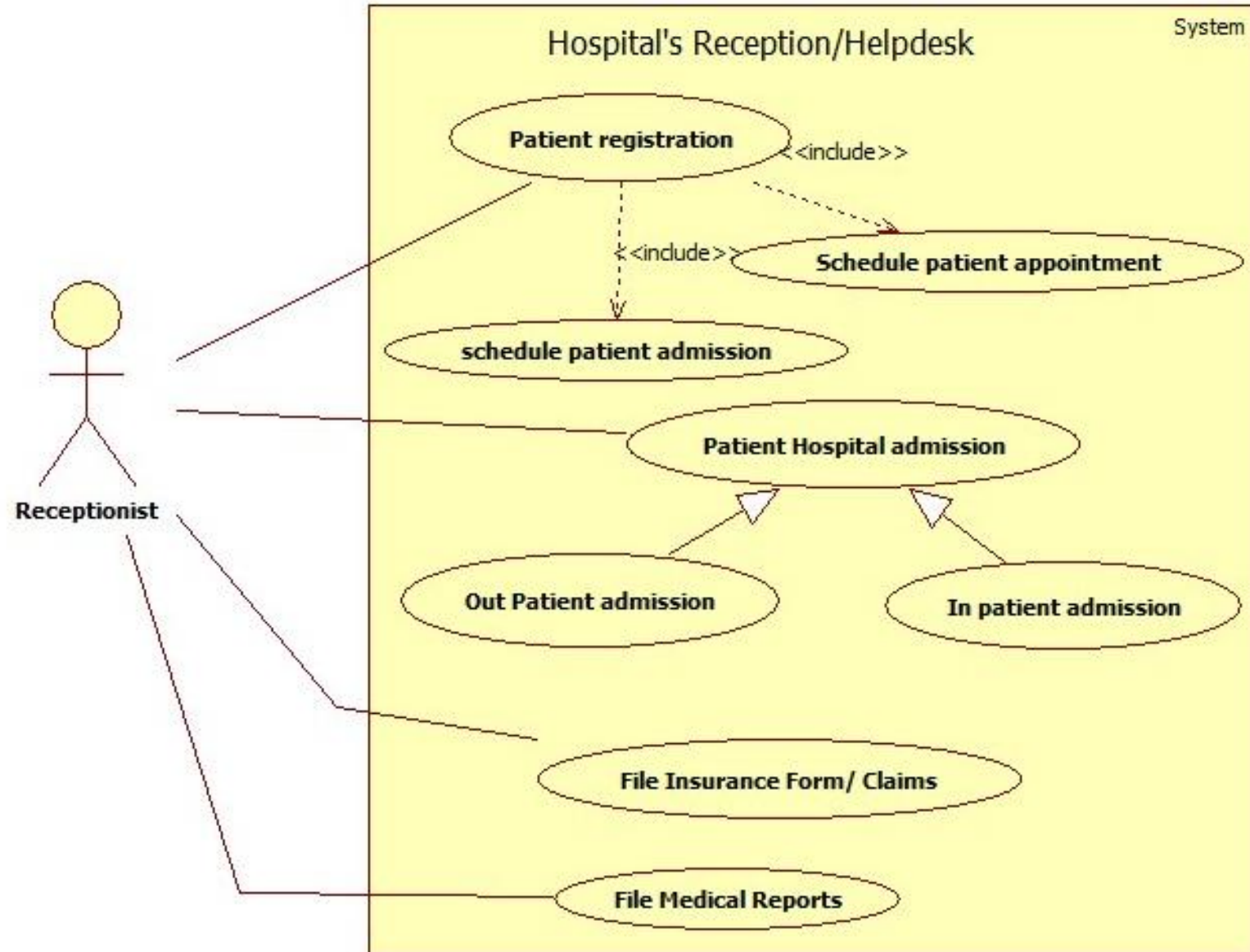
Draw an Use case diagram for the given scenario

Use case diagram for Hospital's reception/helpdesk

- **Purpose:** Describe major services (functionality) provided by a hospital's reception.
- **Summary:** This UML use case example shows actor and use cases for a hospital's reception desk.
 - **Hospital Reception** subsystem or module supports some of the many job duties of a hospital receptionist.
 - Receptionist schedules patient's appointment and admission to the hospital, collects information from the patient by phone and/or upon patient's arrival to the hospital.
- For the patient that will stay in the hospital ("inpatient") she or he should have a bed allotted in a ward. Receptionists might also receive patient's payments, record them in a database and provide receipts, file insurance claims and medical reports.

Use case diagram with advanced relationships

Hospital's reception/ helpdesk





THANK YOU

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